

CO4 – AT3

Course Code: MLA0201

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Constraint-Based Problem Solving

Question 1: Reinforcement Learning for Warehouse Robot Navigation

Problem Understanding

The warehouse robot must reach its destination while avoiding obstacles, minimizing travel time, and conserving battery power. It can move only **Up, Down, Left, and Right**, each movement consumes one battery unit, and the destination must be reached within **25 moves**.

Constraint Analysis

- Obstacles cannot be crossed.
- Maximum allowed moves: **25**.
- Battery capacity: **30 units**.
- Each movement has a battery cost.
- Blocked paths receive penalties.
- The environment is stochastic, so actions may sometimes produce unexpected movement.

Solution Development

Value Iteration with an ϵ -greedy exploration strategy is suitable. Value Iteration calculates the expected value of each state and identifies the policy that maximizes long-term rewards.

The reward system can be designed as:

- Reaching destination: **+100**
- Valid movement: **-1**
- Blocked path: **-10**
- Battery-efficient path: higher overall reward

Concept Application

Value Iteration repeatedly updates the value of each state using:

$$V(s) = \max_a \sum P(s'|s,a)[R + \gamma V(s')]$$

This allows the robot to consider both rewards and uncertainty in the environment.

Justification and Decision

Value Iteration is recommended because it can find an optimal navigation policy while considering stochastic actions. The reward penalties encourage shorter paths, helping the robot satisfy both the **25-move limit** and battery constraints.

Question 2: Multi-Armed Bandit for Online Advertisement Selection

Problem Understanding

The company has **five advertisements** and must select only one advertisement for each visitor. Since the click probabilities are unknown, the system must learn which advertisement performs best while maximizing total clicks within **10,000 interactions**.

Constraint Analysis

- Number of advertisements: **5 arms**.
- One advertisement per visitor.
- Click probabilities are initially unknown.
- Maximum interactions: **10,000**.
- Exploration and exploitation must be balanced.
- Cumulative regret should be minimized.

Solution Development

The problem can be formulated as a **5-Armed Bandit problem**. **Upper Confidence Bound (UCB)** is recommended as the exploration strategy.

UCB selects advertisements using both:

- Average reward received.
- Uncertainty about the advertisement.

Advertisements with fewer observations are explored, while high-performing advertisements are increasingly selected.

Reward Mechanism

The reward is:

- **1** → User clicks the advertisement.
- **0** → User does not click.

The system updates the estimated reward after every interaction.

Justification and Decision

UCB is suitable because it automatically balances exploration and exploitation without requiring a fixed exploration probability. With a limited budget of **10,000 interactions**, it helps reduce cumulative regret and gradually focuses on advertisements with higher click-through rates.

Question 3: Sampling Strategy for Disease Prediction Model

Problem Understanding

The healthcare company has **one million patient records**, but only **8 GB RAM** is available. The selected sample must represent all patient categories while maintaining prediction accuracy above **95%** and model confidence of at least **95%**.

Constraint Analysis

- Dataset size: **1,000,000 records**.
- Memory limit: **8 GB RAM**.
- All patient categories must be represented.
- Required accuracy: **above 95%**.
- Required confidence: **95%**.
- Sampling bias and computational cost must be minimized.

Solution Development

Stratified Sampling is the most suitable method. The dataset is divided into groups based on important patient categories, and samples are selected proportionally from each group.

This ensures that minority categories are not ignored.

Application of Concepts

Sample Complexity: Determines the number of records required to achieve the desired accuracy and confidence.

VC Dimension: Helps estimate the complexity of the learning model. A highly complex model requires more training data.

Occam's Learning Principle: A simpler model that performs well is preferred because it generally has better generalization.

Accuracy-Confidence Boosting: Additional samples or improved training can increase confidence while maintaining the required accuracy.

Justification and Decision

Stratified Sampling is recommended because it provides better representation of different patient categories and reduces sampling bias. It also reduces computational requirements compared with training on all one million records, making the solution practical within the **8 GB RAM constraint**.

Overall Conclusion

Question	Recommended Approach	Main Reason
Warehouse Robot	Value Iteration + ϵ -greedy	Handles stochastic navigation and reward optimization
Advertisement Selection	UCB Multi-Armed Bandit	Balances exploration and exploitation efficiently
Healthcare Sampling	Stratified Sampling	Represents all patient categories while reducing computational cost

These solutions were selected by prioritizing the given constraints and applying suitable machine learning concepts to develop practical and feasible solutions.